

ULTA | MIT COURSE LIST

Short list of courses (see course descriptions below)

5.111 Introduction to Solid State Chemistry

7.016 Introductory Biology

8.01 Physics I: Classical Mechanics

6.100A/B Introduction to Computer Science and Programming with Python + Introduction to Computational Thinking and Data Science

6.110 Fundamentals of Programming

6.120 Mathematics for Computer Science

6.121 Introduction to Algorithms

6.122 Design and Analysis of Algorithms

9.00 Introduction to Psychological Science

9.13 The Human Brain

14.01 Principles of Microeconomics

18.S096 Topics In Mathematics With Applications In Finance

18.06 Linear Algebra

6.036 Introduction to Machine Learning

6.034 Artificial Intelligence

18.700 Theoretical Linear Algebra

FUNDAMENTAL SCIENCES

5.111 Introduction to Solid State Chemistry

Principles of Chemical Science provides an introduction to the chemistry of biological, inorganic, and organic molecules. The emphasis is on basic principles of atomic and molecular electronic structure, thermodynamics, acid-base and redox equilibria, chemical kinetics, and catalysis.

7.016 Introductory Biology

Introduction to fundamental principles of biochemistry, molecular biology and genetics for understanding the functions of living systems. Covers examples of the use of chemical biology, the use of genetics in biological discovery, principles of cellular organization and communication, immunology, cancer, and engineering biological systems. In addition, includes 21st-century molecular genetics in understanding human health and therapeutic intervention.

8.01 Physics I: Classical Mechanics

This first course in the physics curriculum introduces classical mechanics. Our goal is to develop a conceptual understanding of the core concepts, a familiarity with the experimental verification of our theoretical laws, and an ability to apply the theoretical framework to describe and predict the motions of bodies. Introduces classical mechanics. Space and time: straight-line kinematics; motion in a plane; forces and static equilibrium; particle dynamics, with force and conservation of momentum; relative inertial frames and non-inertial force; work, potential energy and conservation of energy; kinetic theory and the ideal gas; rigid bodies and rotational dynamics; vibrational motion; conservation of angular momentum; central force motions; fluid mechanics.

COMPUTER SCIENCE

6.100A/B Introduction to Computer Science and Programming with Python + Introduction to Computational Thinking and Data Science

Antirequisite: if you know how to code in Python, consider skipping this course.

Introduction to computer science and programming for students with little or no programming experience. Students develop skills to program and use computational techniques to solve problems. Topics include the notion of computation, Python, simple algorithms and data structures, testing and debugging, and algorithmic complexity. Provides an introduction to using computation to understand real-world phenomena. Topics include plotting, stochastic programs, probability and statistics, random walks, Monte Carlo simulations, modeling data, optimization problems, and clustering.

6.110 Fundamentals of Programming

PREREQUISITES: 6.100A/B or programming experience

Introduces fundamental concepts of programming. Designed to develop skills in applying basic methods from programming languages to abstract problems. Topics include programming and Python basics, computational concepts, software engineering, algorithmic techniques, data types, and recursion. Lab component consists of software design and implementation.

6.120 Mathematics for Computer Science

Elementary discrete mathematics for science and engineering, with a focus on mathematical tools and proof techniques useful in computer science. Topics include logical notation, sets, relations, elementary graph theory, state machines and invariants, induction and proofs by contradiction, recurrences, asymptotic notation, elementary analysis of algorithms, elementary number theory and cryptography, permutations and combinations, counting tools, and discrete probability.

6.121 Introduction to Algorithms

PREREQUISITES: 6.100A/B and 6.120, or mentor approval

Introduction to mathematical modeling of computational problems, as well as common algorithms, algorithmic paradigms, and data structures used to solve these problems. Emphasizes the relationship between algorithms and programming, and introduces basic performance measures and analysis techniques for these problems.

6.122 Design and Analysis of Algorithms

PREREQUISITE: 6.121

Techniques for the design and analysis of efficient algorithms, emphasizing methods useful in practice. Topics include sorting; search trees, heaps, and hashing; divide-and-conquer; dynamic programming; greedy algorithms; amortized analysis; graph algorithms; and shortest paths. Advanced topics may include network flow; computational geometry; number-theoretic algorithms; polynomial and matrix calculations; caching; and parallel computing.

BRAIN AND COGNITIVE SCIENCES

9.00 Introduction to Psychological Science

A survey of the scientific study of human nature, including how the mind works, and how the brain supports the mind. Topics include the mental and neural bases of perception, emotion, learning, memory, cognition, child development, personality, psychopathology, and social interaction. Consideration of how such knowledge relates to debates about nature and nurture, free will, consciousness, human differences, self, and society.

9.13 The Human Brain

PREREQUISITE: Some experience in Neuroscience/Psychology

Surveys the core perceptual and cognitive abilities of the human mind and asks how these are implemented in the brain. Key themes include the functional organization of the cortex, as well as the representations and computations, developmental origins, and degree of functional specificity of particular cortical regions. Emphasizes the methods available in human cognitive neuroscience, and what inferences can and cannot be drawn from each.

ECONOMICS

14.01 Principles of Microeconomics

This introductory undergraduate course covers the fundamentals of microeconomics. Topics include supply and demand, market equilibrium, consumer theory, production and the behavior of firms, monopoly, oligopoly, welfare economics, public goods, and externalities. Introduces microeconomic concepts and analysis, supply and demand analysis, theories of the firm and individual behavior, competition and monopoly, and welfare economics. Applications to problems of current economic policy.

18.S096 Topics In Mathematics With Applications In Finance

PREREQUISITE: 18.700 and mentor approval

The purpose of the class is to expose undergraduate and graduate students to the mathematical concepts and techniques used in the financial industry. Mathematics lectures are mixed with lectures illustrating the corresponding application in the financial industry. MIT mathematicians teach the mathematics part while industry professionals give the lectures on applications in finance.

MACHINE LEARNING

18.06 Linear Algebra

Basic subject on matrix theory and linear algebra, emphasizing topics useful in other disciplines, including systems of equations, vector spaces, determinants, eigenvalues, singular value decomposition, and positive definite matrices. Applications to least-squares approximations, stability of differential equations, networks, Fourier transforms, and Markov processes. Uses linear algebra software. Compared with 18.700, more emphasis on matrix algorithms and applications.

6.036 Introduction to Machine Learning

PREREQUISITES: Linear Algebra + programming experience

This course introduces principles, algorithms, and applications of machine learning from the point of view of modeling and prediction. It includes formulation of learning problems and concepts of representation, over-fitting, and generalization. These concepts are exercised in supervised learning and reinforcement learning, with applications to images and to temporal sequences.

6.034 Artificial Intelligence (deprecated)

PREREQUISITES: Programming experience

Introduces representations, methods, and architectures used to build applications and to account for human intelligence from a computational point of view. Covers applications of rule chaining, constraint propagation, constrained search, inheritance, statistical inference, and other problem-solving paradigms. Also addresses applications of identification trees, neural nets, genetic algorithms, support-vector machines, boosting, and other learning paradigms. Considers what separates human intelligence from that of other animals.

MATH

18.700 Theoretical Linear Algebra

PREREQUISITE: mathematical experience (talk with Laker)

This course offers a rigorous treatment of linear algebra, including vector spaces, systems of linear equations, bases, linear independence, matrices, determinants, eigenvalues, inner products, quadratic forms, and canonical forms of matrices. Compared with 18.06 Linear Algebra, more emphasis is placed on theory and proofs. Textbook: *Linear Algebra Done Right* by Axler.

Caution: there are no lecture videos for this course. You would read the assigned chapter each week in the book and work through that week's problem set. However, Laker previously taught this course in Spring 2021. If you would like "lecture videos," you can find them on this [unlisted playlist](#) on YouTube. Please do not share publicly.

Information available upon request:

6.S191 Introduction to Deep Learning
8.01 Physics I: Classical Mechanics
8.02 Physics II: Electricity and Magnetism
8.03 Physics III: Vibrations and Waves
8.04 Quantum Physics I
9.40 Introduction to Neural Computation
14.13 Psychology and Economics
18.02 Multivariable Calculus
18.404 Theory of Computation
18.650 Statistics for Applications
22.01 Introduction To Nuclear Engineering And Ionizing Radiation
24.900 Introduction to Linguistics

Short / Non-typical courses:

20.219 Becoming The Next Bill Nye: Writing And Hosting The Educational Show
CMS.116 Creating Video Games
RES.TLL-005 How to Speak
8.20 Introduction To Special Relativity
21A.550J DV Lab: Documenting Science Through Video And New Media